

Context length and readout parameterisation select the emergence exponent of in-context skills in an additive model

[authors]

Draft v0.1 — October 9, 2026

Abstract

We study how a feature-learning model acquires *skills* that are presented in context. Targets are additive models $y = \sum_p c_p \sigma(\langle v_p, x \rangle)$ with mean-zero task coefficients c_p , and a student with learned features and a linear-in-label attention readout must recover the directions v_p by online SGD from prompts of length N . For a single skill of information exponent k^* we obtain the population loss in closed form and derive three consequences. (i) The in-context loss is the in-weight loss with the link squared, so the effective information exponent is $2k^*$ —a fact implicit in one-step analyses of pretraining, which we make dynamical and test by online SGD. (ii) A finite context adds an *alignment-dependent* variance term that makes the unaligned state drift-stable below $m^* \approx \sqrt{2\Gamma/N}$; as a result, the number of prompts cannot substitute for context length, in contrast to the $N \cdot T$ trade-off of one-step analyses. (iii) A trainable scalar readout relaxes to a Wiener shrinkage factor that starves feature learning and raises the effective exponent towards $4k^*$ as N decreases, whereas a readout tied to the feature norm does not. A parameter-free population ODE predicts the SGD-measured exponents across readout protocols and context lengths (12/12 cells within 0.5). With many skills of frequencies π_p the dynamics decouple with emergence times $\propto 1/\pi_p$ —twice as steep an ordering by frequency as in-weight learning—and additive compositions of learned skills cost nothing. In small softmax transformers the effect appears as a $\approx 3\times$ per-token handicap of short contexts accompanied by a collapsing readout, not as a trap, and is invisible when the initial feature alignment is large; we report both regimes.

1 Introduction

Theories of feature learning explain *when* a network acquires a direction v in input space in terms of the information exponent k^* of the link function [2]: online SGD needs d^{k^*-1} samples ($d \log d$ for $k^* = 2$). Theories of emergence and scaling laws [9, 10, 14] then sum many such acquisitions over a power-law dictionary of skills. Both lines treat the skill as something learned *in the weights* from a fixed task. Large models, however, acquire much of their behaviour as *in-context* skills: the pretraining data presents a skill through prompts whose labels are generated by a task-dependent coefficient, and the network must learn features that make the skill recoverable from a context of length N . Recent analyses of in-context learning with learned features [5, 7, 11, 13] treat the context length as an *inference-time* resource: it controls how well a trained model estimates the task, and in one-step analyses of pretraining it enters only through concentration, so that more prompts can substitute for longer prompts [13, Remark 3].

We ask a different question: *does the context length, together with how the readout is parameterised, change which exponent governs the acquisition of a skill during pretraining?* We answer it in the smallest setting in which the question is sharp—one or several skills with $k^* = 2$ (even

links, where in-weight learning is easy), a student with features on the sphere and a linear-in-label readout, online SGD—and then test the predictions outside the solvable model.

Contributions. (1) **Closed form.** The population in-context loss of a single feature is $L_A(m) = 1 - 2\Gamma g(m)^2 + \Gamma^2 g(1)[(1 - \frac{1}{N})g(m)^2 + V(m)/N]$, where m is the overlap with the skill direction, $g(m) = \mathbb{E}[\sigma(\langle v, x \rangle)\sigma(\langle w, x \rangle)]$ and $V(m) = \mathbb{E}[\sigma^2 \sigma^2]$ (Prop. 1). As $N \rightarrow \infty$ it is the in-weight loss with $g \mapsto g^2$: the effective exponent is $2k^*$. The m^{2k^*-1} gradient is present inside the one-step analysis of Oko et al. [13, Lemma 21]; we make the statement dynamical and verify it by online SGD, including the $k^* = 2$ case excluded from Ren et al. [14]. (2) **Finite context creates a drift-stable unaligned state** (Prop. 2): the term $\Gamma^2 V(m)/N$ grows with alignment, so for $k^* \geq 2$ the origin is stable below m^* with $m^{*2} = 16\Gamma/(N(8 - 4\Gamma(1 - \frac{1}{N}) - 24\Gamma/N))$. Under SGD the threshold is soft (noise-activated escape), and increasing the number of prompts per step does not remove it (Fig. 2). (3) **The readout parameterisation selects the exponent** (Prop. 3): a free scalar readout relaxes to the Wiener value $\Gamma^*(m) = g^2/((1 - \frac{1}{N})g^2 + V/N)$ and multiplies the feature gradient, giving effective exponent $4k^*$ where the context statistic is noise-dominated ($d \gg N^{1/k^*}$), while a readout tied to the feature norm has norm-independent directional dynamics and stays at $2k^*$. The population ODE predicts the SGD exponents measured at $d = 32$ for three readout protocols and four context lengths (Fig. 1). (4) **Many skills and composition.** With a power-law dictionary the dynamics decouple with $T_p \propto 1/\pi_p$ (measured slope 1.44–1.53 for $\pi_p \propto p^{-1.5}$), twice the in-weight slope; additive compositions of learned skills emerge with the slower skill and cost no extra time, product compositions are not learned. (5) **Transformers.** In a two-layer softmax transformer in the small-alignment regime ($d = 256$, width 32) a context of 16 with $16\times$ the prompts emerges $\approx 3\times$ later than a context of 256 at equal tokens per step, and the readout norm collapses while the skill is unlearned; at $d = 32$, width 256 (large initial alignment) the ordering reverses. We report both as the boundary of the mechanism.

2 Setting

Inputs $x \sim \mathcal{N}(0, I_d)$. A *skill* is a unit vector $v_p \in S^{d-1}$ with link σ ; we use the normalised Hermite polynomials $\sigma_k = \text{He}_k/\sqrt{k!}$, so $\mathbb{E}[\sigma_j(\langle u, x \rangle)\sigma_k(\langle w, x \rangle)] = \delta_{jk}\langle u, w \rangle^k$ for unit u, w and $k^* = k$. A *task* is a coefficient vector c with independent mean-zero entries, $\mathbb{E}c_p^2 = s_p^2$; a *prompt* consists of N pairs (x_i, y_i) with $y = \sum_p c_p \sigma(\langle v_p, x \rangle)$ and a query (x_q, y_q) from the same task. With a dictionary of P skills and single-skill prompts, skill p appears with frequency π_p .

Model A (in-context). Features $\phi_j(x) = \sigma(\langle w_j, x \rangle)$, $w_j \in S^{d-1}$, and a linear-in-label readout

$$\hat{y}_q = \sum_j \Gamma_j \phi_j(x_q) A_j, \quad A_j = \frac{1}{N} \sum_{i=1}^N y_i \phi_j(x_i), \quad (1)$$

the MLP-plus-linear-attention architecture of Oko et al. [13]. We consider three readout protocols: *fixed* $\Gamma_j = \gamma$; *free*, a trainable scalar; and *tied*, $\Gamma_j = \|u_j\|^2$ with $w_j = u_j/\|u_j\|$ (the 2-homogeneous student of Ren et al. [14]). **Model B (in-weight)** is ordinary regression $\hat{y} = \sum_j a_j \phi_j(x)$ on the fixed task $c \equiv a$. Both are trained by online spherical SGD on the square loss with fresh data at every step.

Write $g(m) = \sum_k \alpha_k^2 m^k$ for $\sigma = \sum_k \alpha_k \sigma_k$ and $V(m) = \mathbb{E}[\sigma(\langle v, x \rangle)^2 \sigma(\langle w, x \rangle)^2] = \sum_j \beta_j^2 m^j$ where $\sigma^2 = \sum_j \beta_j \sigma_j$. For σ_2 : $g = m^2$, $V = 1 + 8m^2 + 6m^4$.

3 A single skill

Proposition 1 (Population loss). *For $P = 1$, one feature at overlap $m = \langle w, v \rangle$, readout Γ and $\mathbb{E}c^2 = s^2$,*

$$L_A(m) = s^2 \left[1 - 2\Gamma g(m)^2 + \Gamma^2 g(1) \left(\left(1 - \frac{1}{N}\right) g(m)^2 + \frac{V(m)}{N} \right) \right], \quad L_B(m) = 1 - 2a g(m) + a^2 g(1). \quad (2)$$

Proof sketch. Conditional on the task, $A = c g(m) + \xi/\sqrt{N}$ with $\mathbb{E}\xi = 0$, $\mathbb{E}\xi^2 = c^2(V - g^2)$; x_q is independent of the context, and $\mathbb{E}[\sigma(\langle v, x_q \rangle)\sigma(\langle w, x_q \rangle)] = g(m)$, $\mathbb{E}[\sigma(\langle w, x_q \rangle)^2] = g(1)$. Expanding $\mathbb{E}(y_q - \hat{y}_q)^2$ gives (2). $\square$

We verified (2) by comparing the Monte-Carlo drift $-\mathbb{E}[\partial_w L] \cdot v$ with $-\partial_m L_A$ at $m \in \{0.05, 0.1, 0.2, 0.4\}$ for $k^* \in \{1, 2, 3\}$: all 24 points agree within 2.3 s.e. and within 3% for $m \geq 0.2$ (Appendix).

The link is squared. As $N \rightarrow \infty$, $L_A = L_B[g \mapsto g^2]$. Near $m = 0$ the drift is $4\Gamma s^2 k^* \alpha_{k^*}^4 m^{2k^*-1}$ against $2ak^* \alpha_{k^*}^2 m^{k^*-1}$ for Model B: the label enters the prediction twice—through the context statistic and through the comparison with y_q —and the mean-zero task prior removes the cross terms that would provide a degree- k^* path. (A bias added to A_j restores an in-weight path of exponent k^* for the *mean* task only; cf. Zhang et al. [16] for the linear case.) With the escape heuristics of Ben Arous et al. [2] (drift $\eta C m^{\kappa-1}$, renormalisation drag $-\eta^2 dm/2$) the timescales are $d^{\kappa-1}$ with $\kappa = 2k^*$: for $k^* = 2$, in-weight needs $d \log d$ samples and in-context d^3 .

Proposition 2 (Finite context: a drift-stable unaligned state). *For σ_2 and fixed readout $\Gamma = \gamma$, $-\partial_m L_A = s^2[(8\gamma - 4\gamma^2(1 - \frac{1}{N}) - 24\gamma^2/N)m^3 - 16\gamma^2 m/N]$. The origin is a local minimum of L_A and the drift is repulsive for $m < m^*$, $m^{*2} = 16\gamma/(N(8 - 4\gamma(1 - \frac{1}{N}) - 24\gamma/N)) \approx 2\gamma/(N(1 - \gamma/2))$.*

The variance $V(m)$ of the context statistic *grows* with alignment—an aligned feature has heavy tails that coincide with the teacher’s—so the finite-context noise penalises alignment. A feature initialised at $m_0 \approx d^{-1/2}$ is attracted only if $d \lesssim d^* = N(8 - 4\gamma(1 - \frac{1}{N}) - 24\gamma/N)/(16\gamma)$ (≈ 31 for $\gamma = 1$, $N = 128$). For $k^* = 1$ the repulsion is a constant fraction γ/N of the attraction and is harmless. One-step analyses of pretraining do not see this term because the readout is initialised at $\gamma \propto d^{-k^*}$ [13, Lemma 18], which makes $\Gamma^2 V/N$ negligible; the phenomenon lives where Γ is order one or trained.

Remark 1 (Soft under SGD). *SGD is a diffusion on the sphere; a run started below m^* escapes with a Kramers-type probability $\sim e^{-\Delta L/D_{\text{eff}}}$, $\Delta L = L(m^*) - L(m_0)$, $D_{\text{eff}} = \eta \text{Var}(\partial_m \text{-gradient})/2$. At $\gamma = 1$, $N = 128$: $d = 16$ has $m_0 > m^*$ (no barrier), $d = 32$ sits on the ridge ($\Delta L/D_{\text{eff}} \approx 0.02$), $d = 64$ is trapped ($\Delta L/D_{\text{eff}} \approx 20$). Observed escapes in $3 \cdot 10^5$ steps: 3/3, 1/3, 0/3. Larger batches lower D_{eff} and deepen the trap.*

Proposition 3 (Readout parameterisation). *(a) Free readout. At fixed m the scalar Γ relaxes at rate $\approx 2V/N$ to $\Gamma^*(m) = g^2/(g(1)[(1 - \frac{1}{N})g^2 + V/N])$, and $L_A(m, \Gamma^*) \approx s^2[1 - Ng^4/(g(1)V)]$ for $g^2 \ll V/N$, i.e. the effective exponent is $4k^*$ while $m \ll N^{-1/(2k^*)}$ and $2k^*$ beyond. (b) Tied readout $\Gamma = \rho = \|u\|^2$: the directional drift is $(1 - m^2)[4gg' - \rho(2(1 - \frac{1}{N})gg' + V'/N)]$ —the attraction does not carry ρ —and $\dot{\rho} = -4\rho \partial_\rho L$ keeps ρ small; the effective exponent is $2k^*$ for every N , while the repulsion of Prop. 2 survives.*

The static value Γ^* is the ridge/Wiener optimum that appears in replica analyses of trained models [5, eq. (298)]; our point is its *dynamical* effect: the optimal readout shrinks an untrustworthy context statistic, and the shrunken readout multiplies the feature gradient, so feature learning is

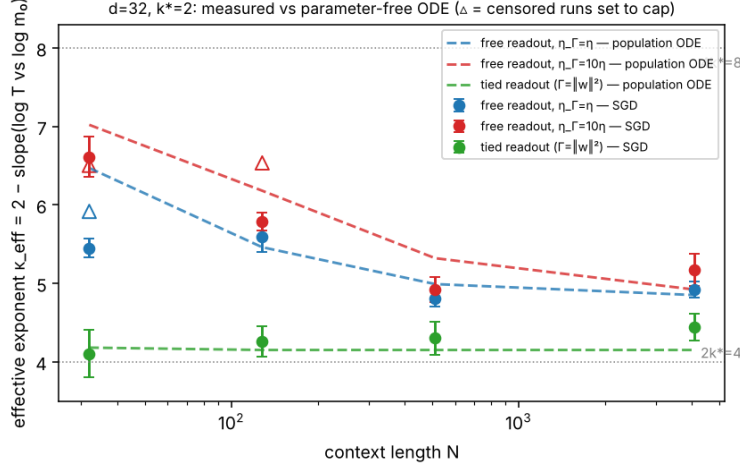


Figure 1: Effective exponent of skill acquisition for $k^* = 2$ at $d = 32$: online SGD (points, $\pm$ s.e.; triangles: censored runs set to the step cap) against the parameter-free population ODE (dashed). The tied readout stays at $2k^* = 4$; the free readout rises towards $4k^* = 8$ as the context shortens.

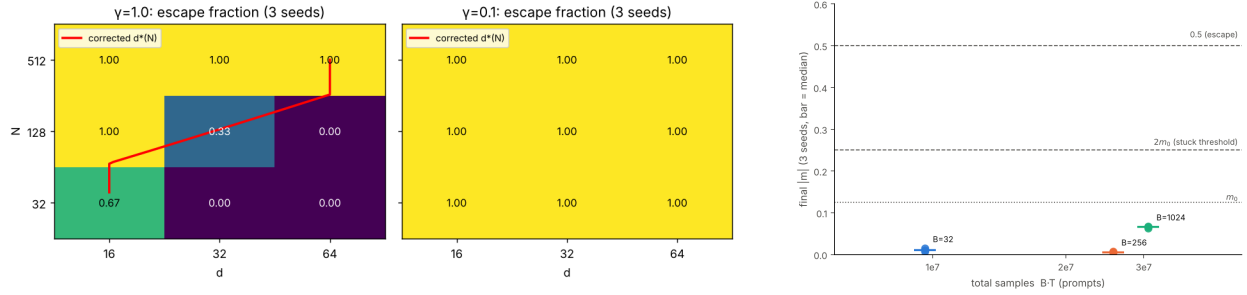


Figure 2: Left: fraction of runs escaping $m_0 = d^{-1/2}$ within $3 \cdot 10^5$ steps (fixed readout; red curve: $d^*(N)$ of Prop. 2). Right: at $\gamma = 1$, $N = 128$, $d = 64$ no run escapes for $B \in \{32, 256, 1024\}$ prompts per step and the final overlap falls below m_0 : more prompts do not substitute for context length.

starved exactly when the context is too short to make the feature useful. That a fast readout doubles the exponent of an in-weight problem is known [4, §5]; what is specific to the in-context problem is that the finite-context noise makes the readout fast, pins it at Γ^* , and makes the outcome depend on whether the readout can shrink independently of the feature norm.

Experiment 1 (exponents by SGD). Online SGD, σ_2 , $d = 32$, $\eta = 1/d^2$, initial overlap $m_0 = c d^{-1/2}$, $c \in \{0.5, 0.7, 1, 1.4\}$, three seeds each; the escape time from m_0 scales as $m_0^{-(\kappa-2)}$ for $\kappa > 2$, so the slope of $\log T_{0.5}$ against $\log m_0$ measures $\kappa_{\text{eff}} = 2 - \text{slope}$. Fig. 1 shows κ_{eff} for free ($\eta_\Gamma = \eta$ and 10η) and tied readouts at $N \in \{32, 128, 512, 4096\}$ against the population ODE of Props. 1–3 evaluated at the same d , m_0 -grid, Γ_0 and η_Γ/η : tied 4.10–4.44 (ODE 4.15–4.18); free 4.8–6.6 rising as N falls (ODE 4.85–7.0); every cell within ≈ 0.2 – 0.5 , the $N = 32$ cells biased low by censoring. The in-weight model at the same k^* gives a logarithmic m_0 -dependence ($\kappa = 2$). The pre-registered asymptotes $4 \rightarrow 8$ are the $d \rightarrow \infty$ limits; at $d = 32$ the same ODE predicts $4.85 \rightarrow 7.0$, which is what SGD shows.

4 Many skills and composition

With P orthogonal skills and $M \geq P$ features the loss splits, when each feature is assigned to one skill, into $\sum_p \pi_p s_p^2 \ell(m_p)$ with ℓ the single-skill loss; each overlap then follows the single-feature dynamics with rate $\propto \pi_p s_p^2 \Gamma$, so $T_p \propto 1/\pi_p$. In-weight learning of the same dictionary carries the coefficient a_p (with $a_p^2 \propto \pi_p$), hence $T_p \propto 1/a_p \propto \pi_p^{-1/2}$: *in-context learning orders skills by frequency twice as steeply*. **Experiment 2** ($d = 32$, $P = 16$, $\pi_p \propto p^{-1.5}$, $M = 64$, fixed $\Gamma = 0.1$, three seeds) gives slopes of $\log T_p$ versus $\log p$ of 1.44 ± 0.08 (drop midpoints; 1.53 ± 0.22 with the 0.5-crossing) for the in-context model and 0.75 ± 0.08 for a capacity-matched in-weight baseline (collective fit; a decoupled 2-homogeneous baseline is reported in the appendix, [\[B-tied numbers\]](#)). The constant $T_p \eta \pi_p m_0^2 = 0.73$ (IQR 0.70–0.77) across skills and seeds equals the single-feature value (0.64), i.e. the many-skill dynamics is the single-feature dynamics with rate π_p . Per-skill drops are $\approx 4\times$ sharper than in-weight (relative width 1.08 vs 4.09). The frequency-weighted alignment loss decays as $t^{-0.40 \pm 0.04}$ over the window in which skills 3–8 emerge, against $(\alpha - 1)/\alpha = 0.33$ for an idealised staircase; the window spans only a factor 3–8 in time, so the exponent depends on the emergence definition and we do not claim a scaling-law ratio.

Composition. Because (1) is linear in the context labels, $\hat{y}_q(c_1, c_2) = \hat{y}_q(c_1, 0) + \hat{y}_q(0, c_2)$ identically, and for independent mean-zero c_1, c_2 the loss on an additive pair is the sum of the single-skill losses. Trained on single-skill prompts only ($d = 32$, $P = 2$, $\pi = 0.75/0.25$, $M = 4$), the pair loss reaches 90% of its drop before or together with the slower skill in every run (never after), in contrast to the extra delay for tuples predicted by Arora and Goyal [1]; the thresholded pair accuracy is approximately the product of the single-skill accuracies ($\text{acc}_{12} \approx 0.85 \text{acc}_1 \text{acc}_2 + 0.09$, r.m.s. 0.008 from $y = x$)—the multiplicative emergence observed empirically by Okawa et al. [12]—while a Gaussian-error closed form fails (heavy tails). A product composition $\sigma_1(\langle v_1, x \rangle) \sigma_1(\langle v_2, x \rangle)$ is never learned (loss ≈ 1).

5 Transformers: where the effect shows and where it does not

We train two-layer softmax transformers (4 heads, pre-LN, Adam, online data) on the same in-context task with $k^* = 2$ and ask whether a short context can be compensated by more prompts per step. **Large initial alignment** ($d = 32$, MLP width 256; the best of 256 random first-layer directions already has $|\cos| = 0.43\text{--}0.55$ with v): at a fixed 8192 tokens per step the emergence step *increases* with N (1000/1400/2800 for $N = 16/64/256$)—fewer prompts per step dominate and the mechanism is invisible, as it must be when $m_0 \gg m^*$. **Small initial alignment** ($d = 256$, width 32; $|\cos|_0 = 0.13\text{--}0.18$): $N = 16$ with $B = 64$ prompts never emerges in 30k steps (3/3); $N = 16$ with $B = 1024$ (the same tokens per step as $N = 256$, $B = 64$) emerges in all seeds but at 10.6–14.8k steps, $\approx 3\times$ later than $N = 256$ (2.8–4.0k); while a run is stuck its final readout norm collapses from 0.57 to 0.03–0.16 and regrows at emergence. Neither pre-registered extreme (a trap, or interchangeability of N and B) holds: in the transformer, context tokens are worth $\approx 3\times$ more than prompt tokens for acquiring an even skill, and the readout shrinks while the skill is unlearned. We state this as an observation consistent with—not a confirmation of—the mechanism of Sec. 3; $k^* = 1$ and in-weight controls emerge within 1.4k and 0.4k steps.

6 Related work

Oko et al. [13] analyse the architecture (1) with one gradient step; the product of two correlations ($\propto m^{2Q-1}$) is in their Lemma 21 and the exponent $2Q$ in their total sample count $N_1 T_1 = \tilde{\Omega}(d^{2Q+1}r)$, framed as a consequence of correlational information; their Remark 3 states the N_1 – T_1 trade-off that Prop. 2 and Fig. 2 contradict beyond the one-step regime. Nishikawa et al. [11] show that softmax’s label nonlinearity lowers the *inference-time* context length to the generative exponent while one-step pretraining stays at $d^{\Theta(\text{ie})}$. Gu et al. [5] derive, by the replica method for trained models with $k^* = 1$, the kernel-vs-feature context-length scalings and the static shrinkage Γ^* ; they list jointly trained readouts and $k^* \geq 2$ as open, which is where our results sit. Kim and Suzuki [7] note that at finite sample size the ICL landscape may acquire a spurious minimum at the origin (their App. C.4), a qualitative precursor of Prop. 2. Ren et al. [14] treat the additive model in-weight for even links with $k^* > 2$ with a 2-homogeneous student; our tied protocol is theirs, and $k^* = 2$ is outside their assumptions. The doubling of the exponent by a fast link is in Bietti et al. [4, §5] and Bietti et al. [3]. Zhang et al. [16] prove, for linear regression, that the task mean is learned in-weight by the MLP and the covariance in-context. Skill-based accounts of emergence [1, 9, 10, 12] are in-weight; our composition result is the label-linear in-context counterpart. Compositions that are not output-additive need extra structure [6, 8, 15].

7 Limitations

The theory is for one feature per skill and $k^* = 2$; the SGD threshold is soft; the many-skill statements rest on decoupling, which fails for rare skills when no feature starts closest to them (2/10 seeds in Exp. 3); the Gaussian closed form for compositional accuracy is wrong; the transformer results are observational and were pre-registered with predictions that failed in both directions. All predictions, outcomes and deviations (including two algebra slips corrected after pre-registration and two ill-posed estimators) are listed verbatim in the appendix.

Reproducibility and disclosure. Code, pre-registration document and all runs: [\[repo\]](#). AI assistance was used for literature search, implementation, derivation checking and drafting; every derivation was verified against the Monte-Carlo drift, and every citation against the full text.

References

- [1] Sanjeev Arora and Anirudh Goyal. A theory for emergence of complex skills in language models. *arXiv preprint arXiv:2307.15936*, 2023.
- [2] Gérard Ben Arous, Reza Gheissari, and Aukosh Jagannath. Online stochastic gradient descent on non-convex losses from high-dimensional inference. *Journal of Machine Learning Research*, 22(106):1–51, 2021.
- [3] Alberto Bietti, Joan Bruna, Clayton Sanford, and Min Jae Song. Learning single-index models with shallow neural networks. In *Advances in Neural Information Processing Systems*, 2022. arXiv:2210.15651.
- [4] Alberto Bietti, Joan Bruna, and Loucas Pillaud-Vivien. On learning Gaussian multi-index models with gradient flow. *arXiv preprint arXiv:2310.19793*, 2023.

- [5] Haotian Gu, Yizhou Xu, and Lenka Zdeborová. In-context learning of single-index targets: Comparing kernel and feature learners. *arXiv preprint arXiv:2610.01712*, 2026.
- [6] Tianyu He, Darshil Doshi, Aritra Das, and Andrey Gromov. Learning to grok: Emergence of in-context learning and skill composition in modular arithmetic tasks. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2406.02550.
- [7] Juno Kim and Taiji Suzuki. Transformers learn nonlinear features in context: Nonconvex mean-field dynamics on the attention landscape. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. arXiv:2402.01258.
- [8] Seijin Kobayashi, Simon Schug, Yassir Akram, Florian Redhardt, Johannes von Oswald, Razvan Pascanu, Guillaume Lajoie, and João Sacramento. When can transformers compositionally generalize in-context? In *ICML 2024 Workshop on Next Generation of Sequence Modeling Architectures*, 2024. arXiv:2407.12275.
- [9] Eric J. Michaud, Ziming Liu, Uzay Girit, and Max Tegmark. The quantization model of neural scaling. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2303.13506.
- [10] Yoonsoo Nam, Nayara Fonseca, Seok Hyeong Lee, Chris Mingard, and Ard A. Louis. An exactly solvable model for emergence and scaling laws in the multitask sparse parity problem. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2404.17563.
- [11] Naoki Nishikawa, Yujin Song, Kazusato Oko, Denny Wu, and Taiji Suzuki. Nonlinear transformers can perform inference-time feature learning. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, 2025.
- [12] Maya Okawa, Ekdeep Singh Lubana, Robert P. Dick, and Hidenori Tanaka. Compositional abilities emerge multiplicatively: Exploring diffusion models on a synthetic task. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2310.09336.
- [13] Kazusato Oko, Yujin Song, Taiji Suzuki, and Denny Wu. Pretrained transformer efficiently learns low-dimensional target functions in-context. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2411.02544.
- [14] Yunwei Ren, Eshaan Nichani, Denny Wu, and Jason D. Lee. Emergence and scaling laws in SGD learning of shallow neural networks. In *Advances in Neural Information Processing Systems*, 2025. arXiv:2504.19983.
- [15] Zixuan Wang, Xingyu Dang, Jason D. Lee, and Kaifeng Lyu. The power of power law: Asymmetry enables compositional reasoning. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. arXiv:2604.22951.
- [16] Ruiqi Zhang, Jingfeng Wu, and Peter L. Bartlett. In-context learning of a linear transformer block: Benefits of the MLP component and one-step GD initialization. *arXiv preprint arXiv:2402.14951*, 2024.

A Derivations

Hermite algebra. For unit u, w with $\langle u, w \rangle = m$ and $z_u = \langle u, x \rangle$, $z_w = \langle w, x \rangle$ jointly Gaussian with correlation m , $\mathbb{E}[\sigma_j(z_u)\sigma_k(z_w)] = \delta_{jk}m^k$. For $\sigma = \sigma_2 = (z^2 - 1)/\sqrt{2}$: $\sigma_2^2 = \sqrt{6}\sigma_4 + 2\sqrt{2}\sigma_2 + 1$, hence $V(m) = \mathbb{E}[\sigma_2(z_u)^2\sigma_2(z_w)^2] = 6m^4 + 8m^2 + 1$ and $g(m) = m^2$. For $\sigma_1 = z$: $g = m$, $V = 1 + 2m^2$.

Population loss (Prop. 1). With $A = \frac{1}{N} \sum_i y_i \sigma(\langle w, x_i \rangle)$ and $y_i = c \sigma(\langle v, x_i \rangle)$, $\mathbb{E}[A | c] = c g(m)$ and $\text{Var}[A | c] = c^2(V(m) - g(m)^2)/N$. Since x_q is independent of the context, $\mathbb{E}[y_q \hat{y}_q] = \Gamma \mathbb{E} c^2 g(m)^2$ and $\mathbb{E}[\hat{y}_q^2] = \Gamma^2 g(1) \mathbb{E} c^2 (g^2 + (V - g^2)/N)$, giving (2).

Drift (Prop. 2). For $\sigma_2, \Gamma = \gamma, s^2 = 1$: $L_A = 1 - 2\gamma m^4 + \gamma^2[(1 - \frac{1}{N})m^4 + (1 + 8m^2 + 6m^4)/N]$, so $-\partial_m L_A = (8\gamma - 4\gamma^2(1 - \frac{1}{N}) - 24\gamma^2/N)m^3 - 16\gamma^2 m/N$, which is negative for $m < m^*$ with $m^{*2} = 16\gamma/(N(8 - 4\gamma(1 - \frac{1}{N}) - 24\gamma/N))$. (An earlier draft of this derivation dropped the $24\gamma^2/N$ term and had a factor 2 in the cubic coefficient; the closed-form loss, which the Monte-Carlo drift check verifies directly, was unaffected.)

Population ODEs (Prop. 3). Spherical gradient flow for the overlap and plain flow for the readout: free readout $\dot{m} = -(1 - m^2)\partial_m L_A$, $\dot{\Gamma} = -r\partial_\Gamma L_A$ with $r = \eta_\Gamma/\eta$; tied readout $\dot{m} = (1 - m^2)[4gg' - \rho(2(1 - \frac{1}{N})gg' + V'/N)]$, $\dot{\rho} = -4\rho\partial_\rho L_A$. The chain rule through $\hat{u} = u/\|u\|$ gives the directional drift a factor $1/\|u\|$ that cancels the $\rho = \|u\|^2$ in the attraction term but not in the repulsion. Escape times $T_{0.5}$ to $m = 0.5$ from $m_0 = c d^{-1/2}$ were integrated with an adaptive step (1% relative change per step) and the exponent read off as $\kappa_{\text{eff}} = 2 - \partial \log T / \partial \log m_0$ over $c \in \{0.5, 0.7, 1, 1.4\}$, exactly as for the SGD data.

Monte-Carlo drift check. $d = 16, N = 128, \gamma = 0.1, a = 1, 5 \cdot 10^4$ batches of 64 prompts per point. Ratios MC/formula at $m = 0.2$ and 0.4 : Model B $k^* = 1, 2, 3$: 0.998–1.00; Model A $k^* = 1$: 1.000; $k^* = 2$: 0.972, 0.998; $k^* = 3$: 0.88 ± 0.76 (drift ≈ 0), 1.02. Model A $k^* = 3$ at $\gamma = 0.1, N = 128$ has negative drift for $m \leq 0.2$.

B Experimental details

All toy experiments: float64 numpy, analytic gradients verified by finite differences (relative error $< 10^{-4}$), online spherical SGD with fresh data each step, $B = 32$ prompts per step unless stated, $\eta = \eta_0/d^2$. Projected samplers (exact in distribution) were used where the gradient depends on x only through a few projections. Compute: Exp. 1 6.4, Exp. 1b 2.8, Exp. 2 3.2, Exp. 3 2.3, Exp. 4 6.0, Exp. 5 6.0 CPU-hours on 4 cores.

Exp. 1 (single skill, uniform initialisation). $k^* \in \{1, 2\}, d \in \{8, \dots, 128\}$, 5 seeds, $N = 128$, fixed $\gamma = 0.1$ (A) and $a = 1$ (B). Slopes of $\log T_{0.5}$ vs $\log d$: B $k^* = 1$: 1.98 ± 0.06 ($d \geq 16$; the pre-registered 1.5 was an error: for $\kappa < 2$ the drift is $O(1)$ and $T \propto 1/\eta = d^2$); B $k^* = 2$: 2.33 ± 0.22 ; A $k^* = 1$: 2.71 ± 0.43 ; A $k^* = 2$: 3.1–3.5 (20 of 105 runs censored at $3 \cdot 10^6$ steps). Regression of $\log(T_{0.5}\eta)$ on $\log |m_0|$: A $k^* = 2$: -2.19 ± 0.08 ($\kappa \approx 4.2$, lower bound because of censoring); B $k^* = 2$: -0.57 ± 0.06 (logarithmic, $\kappa = 2$); A $k^* = 1$: -0.50 ± 0.06 ; B $k^* = 1$: -0.13 ± 0.02 .

Exp. 1b (fixed m_0). Threshold grid: $\gamma \in \{1, 0.1\}, N \in \{32, 128, 512\}, d \in \{16, 32, 64\}$, 3 seeds, $3 \cdot 10^5$ steps. Escapes at $\gamma = 1$: $N = 32$: 2/3, 0/3, 0/3; $N = 128$: 3/3, 1/3, 0/3; $N = 512$: 3/3, 3/3, 3/3 (corrected $d^* = 6.8, 30.8, 126.8$). $\gamma = 0.1$: 27/27. P7: $\gamma = 1, N = 128, d = 64, B \in \{32, 256, 1024\}$ with $3 \cdot 10^5, 10^5, 3 \cdot 10^4$ steps: 0/9 escapes, final $|m|$ 0.005–0.015 ($B \leq 256$) and 0.063–0.067 ($B = 1024$). Exponent table ($d = 32$, 4 values of m_0 , 3 seeds, $\eta = 1/d^2, \Gamma_0 = 0.01$ or $\rho_0 = 0.01$): in Fig. 1; censored runs ($N = 32$: 3/12 for both free protocols; $N = 128, 10\eta$: 2/12) are all at the smallest m_0 .

Exp. 2 (many skills). $P = 16$, $\pi_p \propto p^{-1.5}$, $M = 64$, $\Gamma = 0.1$, $N = 128$, 3 seeds (plus 2 extra and $\alpha = 1$). Skills 1–8 learned in every run; emergence of skills 9–16 partial. In-weight baseline: readout $\rho = \sum_p a_p / M$ (the readout-1 baseline does not learn), collective fit. Slopes and exponents as in Sec. 4; the $\alpha = 1$ slope is 0.88 ± 0.09 (A) vs 0.61 ± 0.08 (B).

Exp. 3 (composition). $P = 2$, $\pi = (0.75, 0.25)$, $M = 4$, $\Gamma = 0.1$, $N = 128$, 5 pre-registered seeds + 5 extra + 3 at $\Gamma = 0.3$. Additivity residual with a paired estimator (same inputs, 17 queries per context): $\max |R| \leq 0.0086$ over 13 runs (the pre-registered independent-set estimator was too noisy, 0.04–0.35). Final neuron assignments (skill 1, skill 2): $(4, 0), (2, 2), (2, 2), (2, 1), (2, 2)$ for the pre-registered seeds. Neuron arrival times $T \approx K / (\eta \pi_p m_0^2)$ with $K = 0.60$ (0.54–0.78).

Exps. 4–5 (transformers). Tokens $[x_i; y_i]$, query $[x_q; 0]$; embedding $\text{Linear}(d+1 \rightarrow 64)$, 2 pre-LN blocks (4-head causal softmax attention + ReLU MLP), linear readout; Adam 10^{-3} , fp32, online data, eval every 200 steps on 2048 held-out prompts, T_e = first eval with normalised MSE < 0.5 . Exp. 4: MLP width 256, $d \in \{16, 32\}$, $N \cdot B = 8192$. Exp. 5: $d = 256$, width 32, cells (a) $N=256, B=64$: $T_e = 4000, 2800$, third seed censored at 17k on the plateau; (c) $N=16, B=64$: none in 30k (3/3); (d) $N=16, B=1024$: 14800, 10600, 11600; (e) $k^* = 1$, $N=16, B=64$: 1400, 1200; (f) in-weight: 400, 200; initial $\max_j |\cos(W_j, v)|$: 0.126, 0.152, 0.176.

C Pre-registration summary

All predictions were written to a version-controlled document before the corresponding run; the full text with timestamps, outcomes and changelog is in the repository (`docs/preregistration.md`). Summary:

ID	prediction	outcome
P1	MC drift of A, $k^*=2$, $\propto m^3$; closed form within 20%	met (slope 3.14 ± 0.07 ; $\leq 3\%$ at $m \geq 0.2$)
P2	$\gamma=1$, $N=128$: stuck for $d \geq 32$	partly ($d=64$ stuck, $d=32$: 1/3 escaped); threshold formula later corrected, $d^* \approx 31$
P3	$\gamma=0.1$: escapes at all d	met (27/27)
P4	free readout: κ rises as N falls, $4 \rightarrow 8$	direction met; magnitudes match the ODE at $d=32$ ($4.85 \rightarrow 7.0$), not the asymptotes
P4'	tied readout: $\kappa \approx 4$ at all N	met (4.10–4.44)
P5	learned $\Gamma \approx \Gamma^*(0.5; N)$	not met (relaxation too slow at $\eta_{\Gamma} = \eta$)
P6	A vs B slope difference ≈ 1 at $k^*=2$	met via m_0 -regression (κ 4.2 vs 2); d -slopes 0.9 ± 0.5
P7	no escape at any B	met (0/9)
P8	transformer, $d=32$: T_e falls $\geq 3\times$ from $N=16$ to 256 at fixed tokens	failed, reversed (1000 $\rightarrow$ 2800)
P9	$k^*=1$ weak N -dependence; in-weight fastest	partly ($\times 2.0$; in-weight fastest)
P10	N -dependence in both softmax and linear attention	met (same direction)
P11a–b	$d=256$, width 32: (c) stuck, (d) stuck	(c) stuck 3/3; (d) emerged 3/3 at $\approx 3\times$ (a): both rival and default failed
P11c	$k^*=1$ and in-weight controls fast	met
P12	additivity residual $ R \leq 0.02$	failed as pre-registered (estimator noise); identity confirmed with a paired estimator
P13	no extra compositional delay	supported with substitute statistic (floor made the original unreachable)
P14	multiplicative accuracy, Gaussian closed form	approximately multiplicative; Gaussian form wrong
P15	product composition not learned	met
P16	T_p slopes 1.5 (A), 0.75 (B)	met (1.44–1.53; 0.75 with capacity-matched B)
P17	A steps $\geq 2\times$ sharper	met ($3.8\times$)
P18	loss exponents 0.33 (A), 0.67 (B)	A within tolerance (0.40); B failed (collective fit saturates)